Residential Demand Response with Power Adjustable and Unadjustable Appliances in Smart Grid

Guodong Zhao[⋆†], Liying Li[♯], Jun Zhang[⋆], and Khaled B. Letaief[⋆]
⋆ Dept. of ECE, The Hong Kong University of Science and Technology, Hong Kong
† National Key Lab. Comm., University of Electronic Science and Technology of China, Chengdu, China
♯ Sch. of Automation Engineering, University of Electronic Science and Technology of China, Chengdu, China

***Abstract*—One major objective in smart grids is to improve the system efficiency by reducing the *peak-to-average ratio* (PAR) of the power consumption, i.e., to balance the supply and demand in the power system. In this paper, we propose to control the power adjustable appliances to reduce PAR in residential buildings. By considering the indoor temperature control appliances, whose operating power is continuously adjustable, we formulate the power management problem as a convex optimization problem to minimize the electricity cost under a comfortable temperature range. An optimal algorithm is developed, which can reduce the cost by about 25% and the PAR by about 63% given non-causal prior knowledge of the power consumption profile of the unadjustable appliances. A practical sub-optimal algorithm that requires only causal knowledge is also developed, which can reduce the cost by about 10% and the PAR by about 28%.**

I. Introduction

A smart grid is an electrical grid with the capability of exchanging measurement and control information between suppliers and consumers through bi-directional communications. This introduces new degrees of freedom on the design and optimization of the grid in real-time. Energy efficiency as well as reliability, economics, etc. can therefore be significantly improved. As a result, smart grids have attracted lots of attention from both academia and industry in recent years [1].

One major objective of the power system is to balance the supply and demand [2]. Otherwise, outage may happen and this will raise serious reliability issues. To avoid such outage, the grid has to be designed for the *peak-demand* rather than the *average-demand*, which, however, leads to under-utilization of the resources and reduces efficiency. With the emergence of the new types of generators and loads, such as solar and wind generators, and *plug-in electric vehicles* (PEVs), both supply and demand become highly dynamic and the *peak-to-average ratio* (PAR) of the power consumption is further enlarged.

Demand response (DR) is an effective way to reduce PAR by scheduling the electricity consumption under *real-time prices* (RTPs). In this case, the retailer usually offers a low price for off-peak hours and a high one for peak hours. Then, the customers may schedule some large-power appliances, called *deferrable loads* [3], such as dishwashers, clothes washers, and clothes dryers, etc., to operate during the off-peak hours. In this way, not only the grid can achieve low PAR, but also the consumers can pay less on electricity bills.

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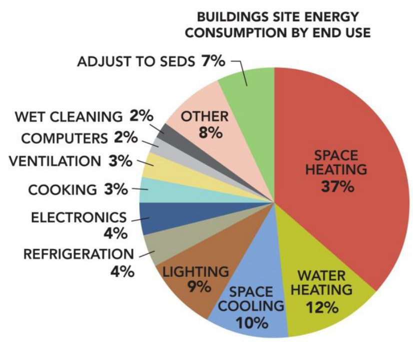

Fig. 1. 2010 U.S. residential energy consumption pie chart. (Source: Section 2.1.1 in [4]. Note: *Wet Cleaning* includes clothes washers, clothes dryers, and dishwashers; *Other* includes small electric devices, heating elements, motors, swimming pool heater, etc.; *Adjust to seds* is the energy adjustment between data sources).

Recently, there were many contributions on DR, e.g., [3], [5]–[8], and the main focus is on the deferrable loads. However, a report [4] released in 2012 from the U.S. *department of energy* (DOE) has indicated that the deferrable loads only take a small share of the overall residential energy consumption, as is shown in Fig. 1. From the figure, the appliances in *wet cleaning* (including clothes washers, clothes dryers, and dishwashers) and *other* (including small electric devices, swimming pool heaters, etc.) sections are deferrable and take less than 8% share. On the other hand, *heating, ventilation, and air conditioning* (HVAC) appliances in *space heating*, *space cooling*, and *water heating* sections take about 72% share and dominate the energy consumption. Thus, managing HVAC appliances is a more efficient way to reduce PAR and increase system efficiency, which is the main motivation of this paper.

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TABLE I
CLASSIFICATION OF RESIDENTIAL APPLIANCES.

Power / Time	Adjustable	Unadjustable
Deferrable	Battery charging, clothes dryer, etc.	Clothes washer, etc.
Undeferrable	HVAC (Heating, ventilation, air conditioning), etc.	Computer, lights, cooking equip. etc.

In this paper, we will investigate the DR on smart HVAC appliances with RTP at one residential site. We aim to minimize the household's electricity cost subject to a comfortable level constraint. In general, all the residential appliances can be divided into different categories, each of which will play different roles in power management. In Table I, from the perspective of the operating time, the appliances are divided into *deferrable* and *undeferrable*, while from the perspective of the power consumption, they can be divided into *adjustable* and *unadjustable*. For example, the kitchen appliances, such as burner and microwave oven, are undeferrable and unadjustable. They are very sensitive to both time and power, and need to be activated immediately with the required power level. This brings the variation of the residential power consumption and results in peak-hours. On the other hand, the temperature control appliances, such as HVAC, are undeferrable, but adjustable*. This is because human beings are not sensitive to a small temperature variation within a comfortable range, e.g., 20.0°C (68°F) to 23.3°C (74°F) in summer [9]. Therefore, our motivation is to reduce the PAR by managing the power adjustable appliances. In particular, we will use HVAC as an example to develop our algorithms, which can be extended to other adjustable appliances.

In summary, the main contributions are as follows,

- Considering both power adjustable and unadjustable appliances as well as RTP, we formulate the residential power management problem as a convex optimization problem with non-causal knowledge of the power consumption profile of *unadjustable appliances* (UAs), in which the electricity cost is minimized subject to a temperature constraint.
- We propose a sub-optimal power management algorithm to control HVAC with the causal knowledge of UA power consumption profiles.
- For the typical residential scenario, we demonstrate the significant performance improvement by our proposed algorithms, where both electricity cost and PAR can be effectively reduced.

II. SYSTEM MODEL

We consider a residential site that is equipped with a smart meter and participates in a RTP program. The smart meter is connected to both an electricity retailer and each appliance via bi-directional communication, which can be realized by either wired or wireless networks [1]. Then, the RTP can be received from the retailer, while the power consumption request information can be collected from appliances. Based on the types of the requests and the current RTP, the smart meter will automatically control the operating power of HVAC.

In the following, we will first introduce the mathematical representations of the power consumption profile shaped by UAs and then provide the indoor temperature model used by HVAC. Finally, we will give the RTP model, which has been widely adopted in recent literatures [10] [11].

A. The Power Consumption Profile of UAs

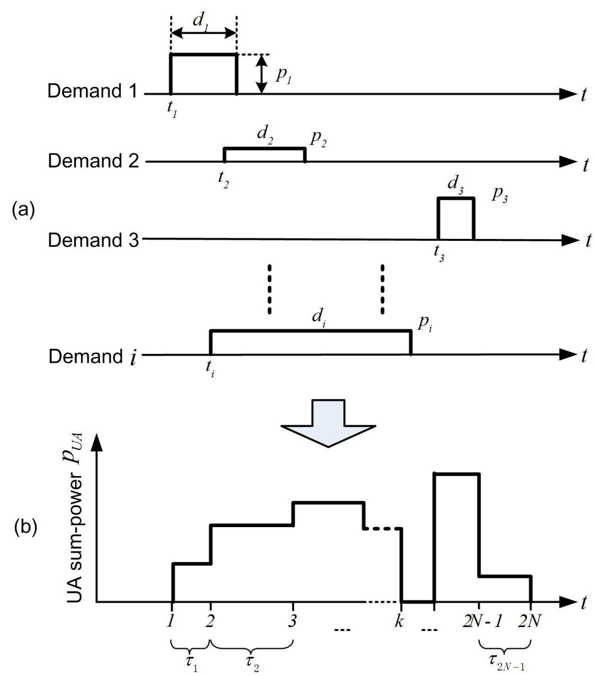

Fig. 2. Power consumption profile shaped by UAs.

Fig. 2 depicts the power consumption profile shaped by UAs. In Fig. 2-(a), the i-th demand is generated at time t_i with an operating power p_i and a service duration d_i. Here, each demand request arrives at the smart meter according to a homogenous Poisson process with an arrival rate λ [3]. We further assume that the operating power p_i of each UA follows a uniform distribution between 0 and $P_{\max}$. In addition, the service duration d_i follows an exponential distribution with a parameter α.

Consequently, the cumulated demands of all UAs shape the power consumption profile as shown in Fig. 2-(b). Then, we have

$$p_{\text{UA}}(t) = \sum_{i \in \Omega(t)} p_i, \tag{1}$$

where $\Omega(t)$ represents the *set* that includes the indices of active UAs at a given time t. From the figure, the sum-power will change at the moment when a UA is turning on or off. With

*Even though most of the current HVACs work under the on-off fashion, there will be more and more inverter HVACs installed in the future, whose operating power is continuously adjustable.

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N demands, there will be $2N$ turning on and off time points, i.e., $k = 1,\ 2,\ \ldots,\ 2N$ in the time axis in Fig. 2-(b). Thus, the sum-power will change $2N$ times, which is corresponding to $2N-1$ time intervals, i.e., $\tau_1, \tau_2,\ \ldots,\ \tau_{2N-1}$.

B. The Temperature Model of HVAC

HVAC is to control the indoor temperature within a comfortable range. When the indoor temperature is different from that of the outdoor, the heat transfer happens. For example, when the indoor temperature is lower than the outdoor in summer, the heat will be transferred from outdoor to indoor, which increases the indoor temperature. To maintain a comfortable temperature, the extra heat needs to be pumped out by air conditioning and ventilation.

Denote P_{trans} (kW) as the heat transfer rate, which is mainly determined by the temperature difference between outdoor and indoor [12]. Here, we assume that P_{trans} is a constant for a given residential site since the small indoor temperature variation will not change P_{trans} too much. Furthermore, we denote $p_{\text{HVAC}}(t)$ and η as the operating power and efficiency of HVAC, respectively[†]. Let $q(t)$ be the indoor temperature at time t, then we have the following temperature model [9],

$$q(t+\tau) = q(t) - \beta\left(\eta \cdot p_{\text{HVAC}}(t) - P_{\text{trans}}\right)\tau, \qquad (2)$$

where β (°C/kwh) is the temperature coefficient, which is the reciprocal of the *thermal house coefficient* [12] and is determined by the size of the residential unit, the structure materials, etc. Thus, β is regarded as a constant for a given residential site.

C. Real-Time Price Model

In smart grid, the RTP is offered by retailers and the motivation is to reduce PAR. Thus, the price usually increases as the power consumption grows, which encourages consumers to consume electricity more constantly. Here, we adopt the following price function from [10] [11],

$$\text{Pri}(n) = a\left(\frac{P_{\text{HVAC}}(t) + p_{\text{UA}}(t)}{C}\right)^b, \qquad (3)$$

where a and b are constants, and C is the capacity of the market.

III. Optimal Algorithm with Non-Causal Profile

Since our algorithm is designed for each residential site, we aim to minimize the electricity cost subject to an indoor comfortable temperature constraint. To obtain the optimal performance, we assume that the smart meter can non-casually obtain all the requests from UAs, including the turn-on time t_i, the operating power p_i, and the service duration d_i.

As indicated before, when a UA is turned on or off, the sum-power $p_{\text{UA}}(t)$ will change accordingly. Then, under the control of the smart meter, the HVAC operating power $p_{\text{HVAC}}(t)$ should be adjusted to a proper value and hold until another turning on or off. Thus, for the UA profile with N demands, we will design $2N$ operating power values, which can be formulated as the following optimization problem

$$\min_{\substack{P_{\text{HVAC}}(n)\\ 0\le n\le 2N}} \sum_{n=1}^{2N} \underbrace{a\left(\frac{P_{\text{HVAC}}(n)+P_{\text{UA}}(n)}{C}\right)^b}_{\text{Price}} \underbrace{\Big(P_{\text{HVAC}}(n)+P_{\text{UA}}(n)\Big)\tau_n}_{\text{Energy consumption}}, \qquad (4)$$

which is equivalent to

$$\min_{P_{\text{HVAC}}(n)} \sum_{n=1}^{2N} \frac{a}{C^b}\left(P_{\text{HVAC}}(n) + P_{\text{UA}}(n)\right)^{b+1}\tau_n, \quad (b > 1) \qquad (5)$$

subject to

$$\sum_{n=1}^{2N}\left(P_{\text{HVAC}}(n)\tau_n\right) = E_{\text{HVAC}}^{\text{total}}, \qquad (6a)$$

$$P_{\text{HVAC}}(n) \ge 0, \quad 0 \le n \le 2N, \qquad (6b)$$

$$Q_{\min} \le q(n) \le Q_{\max}, \quad 0 \le n \le 2N, \qquad (6c)$$

where $E_{\text{HVAC}}^{\text{total}}$ is the total energy consumed by HVAC, $Q_{\min}$ and $Q_{\max}$ are the minimum and maximum comfortable indoor temperatures, respectively.

The first constraint (6a) is the amount of the energy for maintaing the same indoor temperatures before and after the power management. For a given heat transfer rate P_{trans}, there will be $P_{\text{trans}} \sum_{n=1}^{2N} \tau_n$ heat transferred from outdoor to indoor within the $2N$ time intervals. HVAC needs to pump out these heat, so the required energy is

$$E_{\text{HVAC}}^{\text{total}} = \frac{1}{\eta} \cdot P_{\text{trans}} \sum_{n=1}^{2N} \tau_n. \qquad (7)$$

The second constraint (6b) guarantees the non-negative operating power. The third constraint (6c) regulates the indoor temperature variation within a comfortable range, i.e., between $Q_{\min}$ and $Q_{\max}$. By substituting (2) into (6c), we have the following expression with the function of $p_{\text{HVAC}}(n)$,

$$Q_{\min} \le Q_{\text{set}} + \beta\left(\sum_{l=1}^{n}\left(\eta \cdot p_{\text{HVAC}}(l) - P_{\text{trans}}\right)\tau_l\right) \le Q_{\max} \qquad (8)$$

for $0 \le n \le 2N$. This indicates that the operating power of HVAC $p_{\text{HVAC}}(n)$ at each interval also needs to be constrained.

Since all constraints (6a), (6b), and (8) are linear and the objective function is convex, the above optimization problem is convex, which can be solved efficiently. In the simulation section, we will provide its performance as a benchmark for comparison.

IV. Sub-Optimal Algorithm with Causal Profile

The optimal power management algorithm requires the UA power consumption profile. However, such non-causal information is not available to the smart meter. Moreover, it is difficult to estimate the UA profile due to the large uncertainty of human behavior. To implement our idea, we will propose

[†]In the US, the DOE specifies the minimum value of the efficiency η to be 3.43 for new residential HVAC installation after January 2006 [4]. E.g., when $p_{\text{HVAC}}(t) = 1$ kW, the minimum amount of heat that HVAC can pump out for cooling is η kW.

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a sub-optimal algorithm based on the causal profile in this section.

A. Basic Principle

In practice, for most of the high-power UAs, their operating power p_i as well as the service duration d_i can be determined when they are turned on. Taking the microwave oven as an example, people usually set a timer and a required power level, and then turn it on. Therefore, it is possible for the smart meter to collect such information once a UA is turned on. Based on the knowledge of the current operating UAs and their remaining service durations, the causal UA profile can be obtained and the operating power of HVAC can be optimized by solving (5)-(6c). Then, the smart meter will control HVAC. In the coming time interval, if another UA is turned on, the UA profile will be updated and the smart meter will recalculate the operating power of HVAC for the coming interval.

B. Algorithm Description

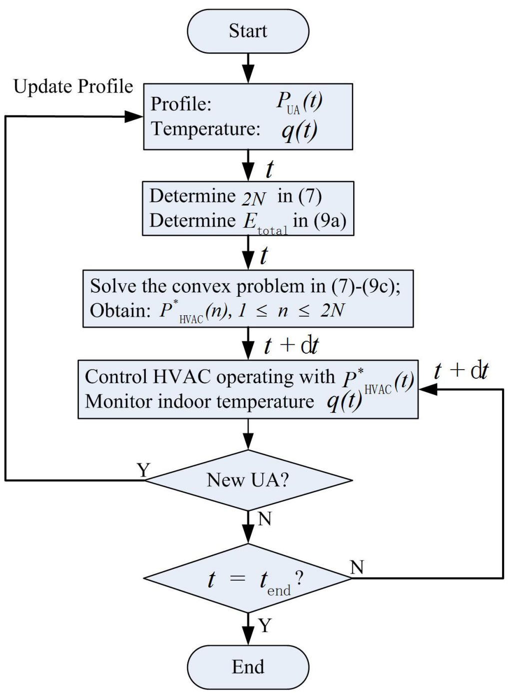

Fig. 3. Flow diagram of the sub-optimal algorithm with causal profile.

Fig. 3 shows the flow diagram of the algorithm with the causal profile information. The HVAC will be controlled by the smart meter according to the following steps:

1) At time t, calculate the UA power consumption profile according to current operating UAs as well as the ending time of HVAC power management, i.e., $t_{\text{end}} = t + \sum_{k=1}^{2N} \tau_k$. Meanwhile, the smart meter collects the initial indoor temperature $q(t)$ from HVAC.
2) Determine the number of intervals $2N$ based on the UA power consumption profile, i.e., τ_1, τ_2, ..., τ_{2N}. Then, the required energy of HVAC $E_{\text{HVAC}}^{\text{total}}$ can be calculated by (9), which will be described at the end of this subsection.
3) Solve the convex optimization problem in (5)-(6c) and obtain the operating power $p_{\text{HVAC}}^*(n)$ for $1 \le n \le 2N$.
4) Control HVAC with $p_{\text{HVAC}}(t) = p_{\text{HVAC}}^*(n)$ for $t \in \tau_n$.
5) Monitor the indoor temperature as well as new active UAs. If no UA is activated, the smart meter will continue to control HVAC as in Step 4. Otherwise, if a UA is activated, the smart meter will update the casual profile as well as the initial indoor temperature $q_0(t)$, and then re-calculate the operating power for the rest of time.

In Step 2, the energy required for HVAC can be obtained by the following expression

$$E_{\text{HVAC}}^{\text{total}} = \underbrace{\frac{1}{\eta} \cdot P_{\text{trans}} \sum_{n=1}^{2N} \tau_n}_{\text{Maintenance}} + \underbrace{\frac{q_0(t) - Q_{\text{set}}}{\eta\beta}}_{\text{Adjustment}}. \tag{9}$$

The first term is the energy required for maintaining the indoor temperature, i.e., after the HVAC power management, the temperature will return to the initial value $q_0(t)$. The second term is the energy adjustment, which forces the temperature to return to the preset value Q_{set}. When $q_0(t) > Q_{\text{set}}$, the second term is positive, which means that HVAC needs more energy to let the temperature return to Q_{set}. Otherwise, when $q_0(t) < Q_{\text{set}}$, less energy is required.

In Step 3, the optimization problem can be efficiently solved by multi-level water filling. Specifically, we omit the temperature constraint (6c) and obtain a simple expression in water-filling fashion, i.e.,

$$P_{\text{HVAC}}^*(n) = \left(C \left(\frac{-\mu^*}{a\tau_n} \right)^{\frac{1}{b}} - P_{\text{UA}}(n) \right)^+, \tag{10}$$

where $(x)^+ \triangleq \max(0, x)$ and μ^* is the water level chosen to satisfy the energy constraint with equality (6a). Then, for $2N$ power values, we will check whether each of them satisfies the temperature constraint (6c) in an increasing order, i.e., from $P_{\text{HVAC}}(1)$ to $P_{\text{HVAC}}(2N)$. If so, we will check the next one, otherwise, we will re-calculate the power values for the rest intervals with the remaining energy.

V. Simulation Results

In this section, we will demonstrate the performance of our proposed algorithms. We assume that the number of activated UAs within the unit duration $T = 1$ hour follows a Poisson distribution with $\lambda = 2$, the operating power follows a uniform distribution between 0 kW and 3 kW, and the average service duration follows an experiential distribution with $\alpha = 1$, i.e., the average service duration is 1 hour. For the RTP function (3), we assume that $a = 1$, $b = 3$, and $C = 1$. We further assume that our considered residential site is of 150 square meters, the temperature difference between indoor and outdoor is 10°C, then the heat transfer ratio is $P_{\text{trans}} = 8.7$ kW according to [12]. In the simulation, the time interval is set to be $\tau_{2N} = 0.5$ hour, the cooling efficiency of HVAC is $\eta = 3.43$ according to Table 5.2.4 in [4], the preset indoor temperature Q_{set} is 22°C, and each curve is obtained under 200

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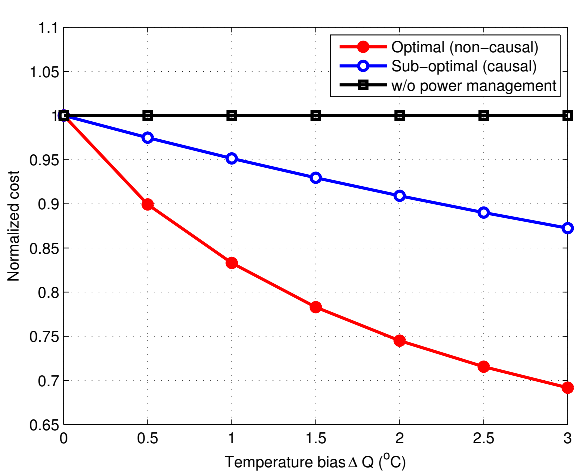

Fig. 4. Electricity cost versus the temperature bias.

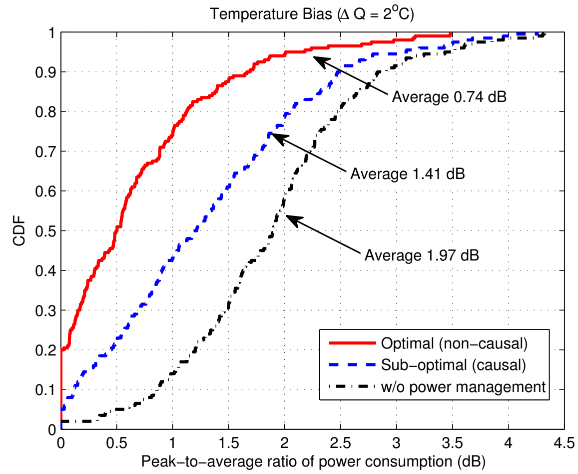

Fig. 5. PAR reduction of different algorithms.

monte carlo trials. We denote ΔQ as the indoor temperature bias, which indicates the allowable variation range of the indoor temperature, then we have $Q_{\min} = Q_{\text{set}} - \Delta Q$ and $Q_{\max} = Q_{\text{set}} + \Delta Q$.

A. Normalized Cost v.s. Indoor Temperature Bias

Fig. 4 shows the normalized electricity cost versus the indoor temperature bias, where both the optimal and sub-optimal algorithms are considered. We also provide the performance of the method without power management for comparison, the normalized cost of which is always equal to 1, i.e., the highest cost. This is because HVAC operates with a constant power and has nothing to do with the temperature bias. For the optimal algorithm, the cost reduces dramatically as the temperature bias ΔQ grows. We also observe a similar trend on the sub-optimal algorithm. In particular, for a typical comfortable range, where $Q_{\text{set}} = 22°\text{C}$ and $\Delta Q = 2°\text{C}$ [9], the optimal and sub-optimal algorithms can reduce costs by about 25% and 10%, respectively.

The performance loss of the sub-optimal algorithm is because that the optimal one has non-causal knowledge of the power consumption profile of all UAs, and then the operating power can be optimized across the whole profile. Whereas, the sub-optimal one has only the causal profile of current active UAs, i.e., partial profile, and then the optimization has to be re-calculated once a new UA is active.

B. CDF Curve of PAR

Fig. 5 compares the *cumulative distribution function* (CDF) curves of different algorithms, where the temperature bias is $\Delta Q = 2°\text{C}$. We see that the power consumption PAR can be effectively reduced by the proposed algorithms, i.e., the total power consumption becomes more stable. In particular, the optimal algorithm with non-causal knowledge of the UA profile reduces PAR by about 63% on average (from 1.95 dB to 0.74 dB), while the sub-optimal one with the causal knowledge reduces the PAR by about 28% on average (from 1.95 dB to 1.41 dB).

VI. Conclusions

In this paper, we investigated the residential demand response considering not only the power adjustable but also the power unadjustalbe appliances. In particular, the operating power of the former ones was optimized according to the usage of the latter ones. As a result, the overall power consumption becomes more stable and the electricity cost is minimized. We found that the non-causal knowledge is critical and predicating UA profile can potentially improve the performance. Simulations also indicated that the proposed algorithms can effectively reduce both the electricity cost and the PAR effectively.

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